

AKSHAR VANDARA

Boston, MA | (857) 919-6234 | axr230102@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)
Software/Research Engineer • 3 peer-reviewed publications in PCG and constraint optimization

EDUCATION

Northeastern University — *M.S., Game Science and Design*
GPA: 3.87 / 4.0 | Coursework: Game Data Science, Game AI

May '26
Boston, MA

Gujarat Technological University — *B.E., Computer Engineering*
GPA: 3.6 / 4.0 | Coursework: Algorithms, AI, ML

May '23
Ahmedabad, India

SKILLS

Languages: C#, C++, Python, Java, R, C, SQL, JavaScript
Game Dev: Unity (2D/3D), Unreal Engine, Blueprints, Behavior Trees, EQS, Perforce, ParaView/VTK
ML / AI: PyTorch, TensorFlow, NumPy, Pandas, CUDA, Z3 SMT, PySAT, SAT/CSP modeling, OpenCV
Backend / Web: Firebase, Flask, AWS, REST APIs, GCP, React, Django, Node.js,

EXPERIENCE

Research Assistant

Feb '25 – Present

Game Design Lab, Northeastern University

Boston, MA

- Engineered constraint-solving level-generation pipeline called **Sturgeon (PySAT)** that generates 2D puzzle levels with valid, solvable paths, cutting generation time **100–500×** vs. Spacetime WaveFunctionCollapse (*first-author, AAAI AIIDE 2025*).
- Designed an interactive editor for real-time level/path editing, plus **3D voxel visualization** via VTK/ParaView.
- Extended the system to **4D spacetime blending** of learned game mechanics (*first-author, FDG 2026 PCG Workshop*).
- Built a Python constraint solver (**Z3 SMT**) for generating **continuous spacetime** gameplay (*first-author, FDG 2026 PCG Workshop*).

Gameplay Systems & UI Programmer

May '25 – Aug '25

Otisco Studios

Orlando, FL

- Programmed **8+ core gameplay systems** in Unreal/C++ (including save/load and UI) across two titles.
- Created **5+ NPC** and enemy AI archetypes for roguelike and open-world genres using C++, Behavior Trees, and **custom EQS**.
- Resolved **30+** critical pre-launch bugs to stabilize the build for release.
- Profiled and refactored gameplay code, **doubling average framerates (2x)**.

Unity Developer

Mar '23 – Jul '24

Arcadon Games

Bangalore, India

- Developed core gameplay mechanics in Unity/C# using state machines, improving input **responsiveness 30%**.
- Integrated Firebase to store and sync **1,000+** user profiles and inventory data, with sub-100ms response.
- Wrote a match-simulation system that processes **30 matches/min** and cuts load times 40%.
- Ran alpha testing with **100+ users (85% satisfaction)**; fixed 20+ stability bugs before launch.

PROJECTS

Spacetime PCG Research Suite | [GitHub](#) | [Publication](#) | *Python, Z3 SMT, PySAT, Pandas, Paraview*

- Open-sourced an extension of Sturgeon for spacetime-constrained level generation; supports 2D, 3D-blended, and continuous physics-based platformer domains.
- Benchmarked generation speed and solvability across the three domains with Pandas, surfacing per-domain performance trade-offs.

Red Circuit Rebellion | [Steam](#) | *Unreal Engine 5, C++, Blueprints, Behavior Trees, EQS*

- Shipped a title on Steam with two distinct game modes (Story and Arcade), each with its own objective logic and enemy spawners.
- Implemented standout gameplay features in C++ and Blueprints, including real-time environmental fracturing and a dynamic objectives manager.

Cricket Manager: League | [Playstore](#) | *Unity, C#, Firebase, Cloud Functions*

- Architected sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features.
- Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.